

Yesterday's Count, Tomorrow's Rain

Forecasting vs Naive Restocking of the Campus Loaner-Umbrella Bins

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Background

Wet mornings empty the campus umbrella bins within the hour; dry ones leave them stocked and unused. The Horizon Register forecasts wet-weather demand three days out. Does the forecast actually improve next-day restocking over a caretaker who simply repeats yesterday's count?

Aims

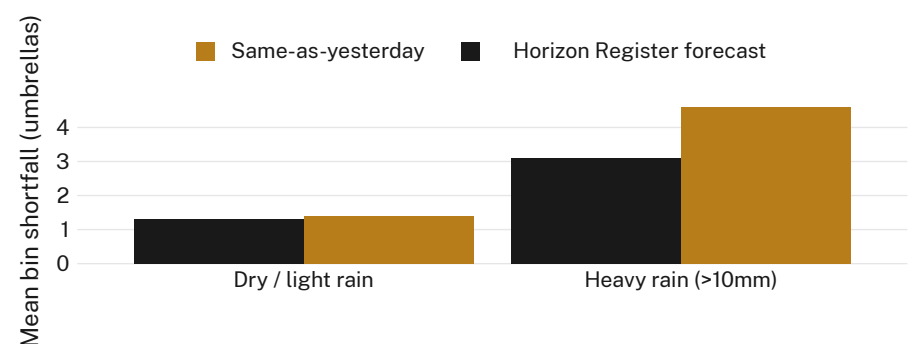
- Compare next-day restocking accuracy under the Register's forecast rule against a same-as-yesterday rule
- Establish whether any forecast advantage holds uniformly or only above a rainfall threshold
- Estimate the caretaking cost of switching rules

Methods

- 6 loaner-umbrella bins across two schools · daily stock counts · 14-week term
- Register forecast probability compared against a persistence baseline (yesterday's closing count)
- Rainfall bands pre-registered; each day's restocking decision logged before the count, never adjusted post hoc

Results

Restocking shortfall under the Register's forecast is indistinguishable from the baseline on dry and light-rain mornings. On days with forecast rainfall above 10mm, the forecast rule cuts mean shortfall by roughly a third.



Mean daily umbrella shortfall on the six monitored bins fell from 4.6 to 3.1 units on forecast-heavy-rain mornings between weeks 3 and 11 of Semester 1, 2026 (n = 84 bin-days); shortfall on drier mornings showed no reliable difference between rules.

Discussion & conclusions

The Register's forecast earns its keep only when it is least convenient to ignore: heavy-rain mornings. On ordinary days it adds no detectable value over a caretaker's memory of yesterday, and the saving surfaces only where the baseline was already failing hardest. Next: extending the audit to the remaining campus bins across a full academic year.

References

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Rainfall bands pre-registered; bin counts logged by rostered caretaking staff; no umbrellas retained or identified.

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